Results

Beginning with the fixed-effects model, this acts as the baseline model, controlling for unobserved heterogeneity across households and time. The model controls for 447 household fixed effects and 11 time periods, with standard errors clustered at the household level. The household fixed-effects control for time-invariant characteristics of a household such as assets, debts, family size and site of experimentation. While the time fixed-effects capture economy-wide shocks that affect hours worked. These fixed affects allow us to isolate the effect of wage changes and GAI payment changes on hours worked. Specifically, the coefficient of HH_WAGE is 0.496, suggesting that a $1 increase in a household's wage is associated with an increase of 0.5 hours worked, ceteris paribus. With a p-value of <0.01 this coefficient is statistically significant at the 1% level, which is strong supporting evidence that higher wages are associated with greater labour supply. Conversely, the model suggests that an increased GAI payment is associated with a decrease in household hours worked by about 0.62 hours. This coefficient is statistically significant at the 5% level.

The difference-in-differences model builds previous model, comparing the differences before and after the policy intervention for control vs treated groups while accounting for household and time fixed-effects. The coefficient of $TREATED \times POST$ is -108.53 indicating there is an association between receiving GAI payments with a reduction of working hours by approximately 109 hours in a period. Given a period is about 5-6 months (800-960 working hours), a reduction of 109 hours is about 11-14% decrease in working hours. This result is statistically significant at the 5% significance level (p value = 0.0216) meaning the null hypothesis that GAI payments do not effect household hours worked can be rejected. These results are consistent with previous studies such as Hum

and Simpson, who found similar effects, a decrease in working hours, however with much higher magnitude effects.

To capture the effect of cultural values on treatment, specifically, how racial background affects, we consider race as a tertiary factor. If the household contains a person of racialized background, it is considered a racialized household which is then used in the triple interaction term. The coefficient for the $TREATED \times POST$ interaction remains the negative with a similar magnitude and is still significant at the 5% level, suggesting the same results as the previous DiD model. However, the coefficient for the triple difference term ($TREATED \times POST \times RACIALHH$) is not significant at the 5% level suggesting there is insignificant evidence to reject the null hypothesis, i.e. the treatment effect does not differ systematically between racialized and nonracialized households. Overall, the reduction in working hours due to the GAI policy does not significantly vary between the groups.

These results suggest that a guaranteed annual income policy is associated with a reduction in household labour supply. The fixed effects model shows that higher wages increase hours worked, while higher payments decrease hours worked. This is consistent with standard economic theory that non-labour income decreases work incentives. The difference-in-differences model further indicates that the policy intervention led to a decline in households that received payments relative to treated households. However, the triple-difference model suggests there are not significant differences between racialized and nonracialized households when in relation to labour supply under a GAI regime. While the policy affects labour supply, the reduction in hours worked is similar across both groups. From a policy perspective, these results suggest that government income

programs may reduce labour supply broadly, but does not disproportionately affect labour supply across different racial groups. (See appendix A)

Overall Economic Interpretation

The results of empirical analysis suggest that treatment (participation in the GAI program) is associated with a reduction in household labour supply (household hours worked). The fixed-effects model indicates that higher wages are associated with more hours worked while higher payments are acclimated with less hours worked, on average. Specifically, the coefficient on household wages is positive, suggesting that increasing wage increases labour supply, which is consistent with standard labour supply theory which claims higher returns to work increases labour supply incentives. Conversely, the negative coefficient on payments could be a realization of the income effect; the household's real income changes leading to a reduction in working hours. Identification of these variables depends heavily on within-household variation over time that is, household effects and period fixed effects. We assume that there is minimal change in characteristics of a household such as education, preferences, and other unobservable behaviour, which allows the model to control for these factors. Period or time-fixed effects control for common shocks that effect all or most households in a given period. The fixed effects model assumes that changes in payment or wages is exogenous, conditioned on these controlled variables. Anticipation of the policy could lead to anticipation bias, leading to a upward bias, overestimating the true causal impact.

The DiD model builds on the fixed effects model by comparing changes in household working hours between treated and control households before and after the policy is implemented. This requires the parallel trend assumption which assumes that treated and

control groups would have followed similar trends without GAI. Given the limited amount of pre-experiment data, this assumption is difficult to justify. However, by using an event-study (Appendix B) this shows the lag in the effects of GAI, with smaller non-significant effects in earlier periods. This shows the assumption holds unless treated households were already on a declining trend prior to the experiment. This would lead to downward bias attributing pre-existing trends to the policy.

Another salient concern is the presence of external shocks and policy changes due to the 1973 oil crisis. This global event would create macroeconomic instability, rising inflation and labour market fluctuations which could affect treated and control households differently. For example, if treated households were disproportionally concentrated in weaker labour markets, treatment effects may be artificially biased downward, capturing broader economic conditions, rather than the causal effects of the GAI payments. This is because period fixed effects can only control for shocks that effect all households equally.

Specifying the coefficients for the triple-difference model makes the same assumptions as above, however, the insignificance of the term suggests that there is no significant difference in working hours between racialized and non racialized households. This significance could be due to limited sample size or statistical power since racialized households made up a particularly small proportion of the population (about 2%).

As a robustness check, all three models were retrained based on heteroskedacity-robust standard error, meaning we remove the assumption that variance of the error term is constant over all observations. This may be the case for households with stable hours versus those with unreliable or seasonal work. Even without this assumption the models estimate similar coefficients with still statistically significant results. This suggests that the

main results are not sensitive to potential violations of the homoskedacity assumption. (See appendix C)

Policy Implications & Conclusions

The results of this analysis imply that unconditional income programs such as the Mincome experiment have salient impacts on the labour supply. This reaffirms that non-labour income is positively associated with reductions in hours worked while higher wages incentivize households to participate and increase their amount of labour supply in the market. The difference-in-differences model further suggests that increasing mincome payments reduced hours worked through the income effect. These findings show the impact of guaranteed income programs on labour market behaviour, reducing the need for households to rely on labour income. This contributes to ongoing policy decisions regarding universal basic income and other income programs by highlighting and measuring the potential trade-off between income support and labour market participation.

It should be noted that the decrease in labour supply are not necessarily implicative of negative welfare outcomes. A reduction in work hours should be interpreted within context. Income programs allow households to allocate time away from work and towards leisure activities such as education, caregiving, or job search. The lack of significant difference between treatment effects of racialized and non racialized households suggests that effects were broadly similar across racial lines. Policymakers must weigh potential reductions in labour supply against benefits of improved financial security and reduced poverty. If the latter outweighs the former, households may accept the policy decision, increasing economic stability with small changes in labour supply.

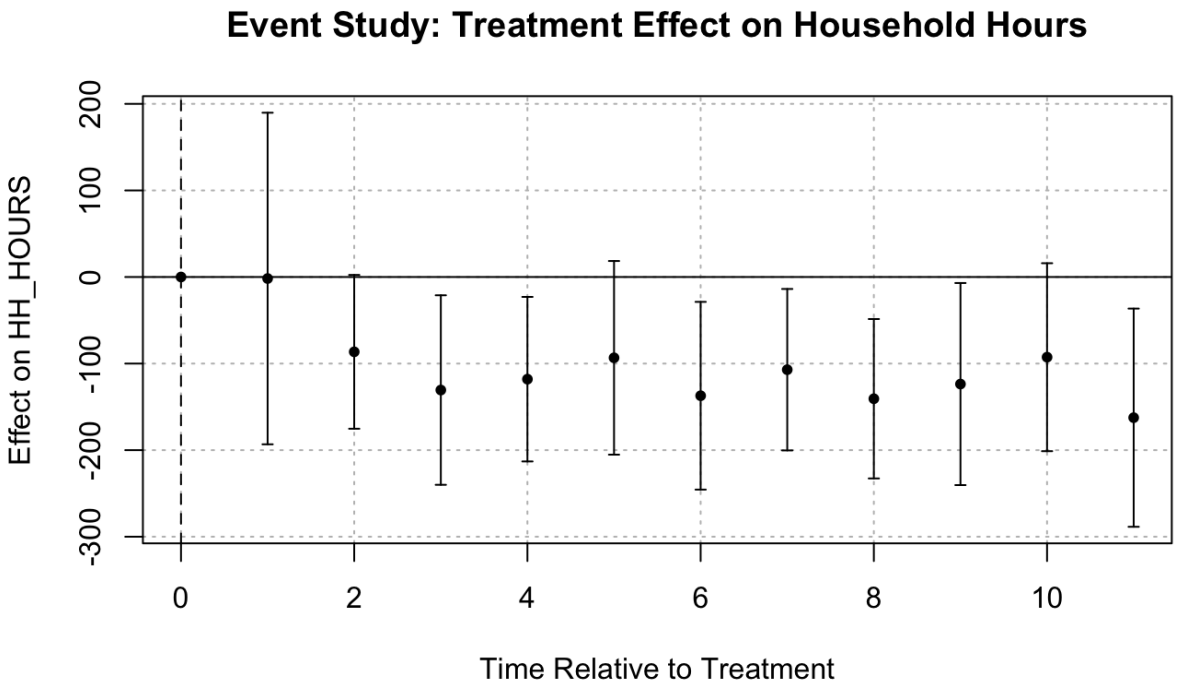
These results provide empirical evidence that unconditional income for households can reduce their labour supply, with similar effects across racialized and nonracialized households in the sample. Although there are risks, broader welfare implications depend on policymakers' ability to balance labour market participation with socially beneficial goals that increase social welfare.

Appendix A

	Fixed Effect Model	Diff-in Diff Model	DDD Model
HH_WAGE	0.496***		
	(0.107)		
PAYMENT	−0.616*		
	(0.245)		
TREATED × post		−108.525*	
		(47.078)	
treat_post			−114.822*
			(52.375)
triple_diff			−33.526
			(63.335)
Num.Obs.	4917	5168	5168
R2	0.639	0.589	0.589
R2 Adj.	0.602	0.548	0.549
R2 Within	0.128	0.001	0.003
R2 Within Adj.	0.127	0.001	0.002
AIC	70 244.3	74 430.5	74 422.4
BIC	73 228.0	77 443.6	77 442.0
RMSE	278.86	296.83	296.53
Std.Errors	by: FAMNUM	by: FAMNUM	by: FAMNUM
FE: FAMNUM	X	X	X
FE: period	X	X	X

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Appendix B


Event Study: Treatment Effect on Household Hours
Effect on HH_HOURS
200
100
0
-100
-200
-300
0
2
4
6
8
10
Time Relative to Treatment

Appendix C

	FE Robust	DiD Robust	DDD Robust
HH_WAGE	0.496***		
	(0.094)		
PAYMENT	−0.616**		
	(0.205)		
TREATED × post		−118.011*	−114.822*
		(53.091)	(53.570)
TREATED × post × RACIALHH			−33.526
			(69.082)
Num.Obs.	4917	5168	5168
R2	0.639	0.589	0.589
R2 Adj.	0.602	0.549	0.549
R2 Within	0.128	0.003	0.003
R2 Within Adj.	0.127	0.002	0.002
AIC	70 244.3	74 420.9	74 422.4
BIC	73 228.0	77 434.0	77 442.0
RMSE	278.86	296.55	296.53
Std.Errors	Heteroskedasticity-robust	Heteroskedasticity-robust	Heteroskedasticity-robust
FE: FAMNUM	X	X	X
FE: period	X	X	X

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001